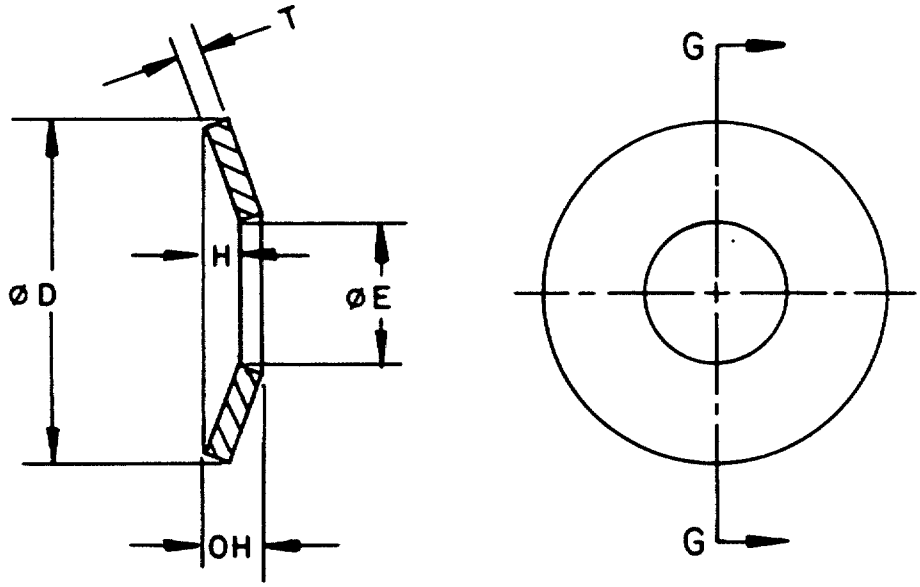
METRIC
DoD-W-63474/2A
18 April 1986
SUPERSEDING
DoD-W-63474/2
4 May 1984

MILITARY SPECIFICATION SHEET

WASHER, SPRING TENSION,
(BELLEVILLE SPRING), METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the washers described herein shall consist of this document and the latest issue of DOD-W-63474.



SECTION G-G

NOTE:
1. All dimensions are in millimeters, see Table I for dash numbers and dimensions.

FIGURE 1. WASHER, SPRING TENSION, (BELLEVILLE SPRING), METRIC

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. Material: Spring steel, Grades 1050 thru 1095 in accordance with ASTM A684/A684M and alloy steel, Type 6150 (UNS G61500) in accordance with AMS 6455.

2. Surface hardness: Spring steel washers shall have a surface hardness of 43-50HRC (82-85.5 HR15N for thickness less than 1mm), alloy steel washers shall have a surface hardness of 45-52HRC.

3. Protective coating and surface treatment: Plain or uncoated and cadmium plate.

4. Part number: The part number consists of the letter "D" the basic number of this specification sheet, the finish code letter, and a dash number taken from Table I.

Example: D63474/2-P030

- Dash number
- Finish codes
  - P - Plain or uncoated
  - C - Cadmium plate
- Specification sheet number

D63474/2-P030 indicates: Washer, spring tension (Belleville spring); Spring steel; Plain or uncoated; 6.2 mm ID; 12.5 mm OD; and 0.7 mm thick.

5. Revision letters are not used to denote changes due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

TABLE I. DASH NUMBERS AND CHARACTERISTICS.

DASH NO	ØE ID	TOL	ØD OD	TOL	T THICKNESS	TOL	CH	TOL	H FREE HEIGHT	TOL	LOAD NEWTONS	TOL %	DEFLECTION 75% H (APPROX)
010	4.2		8		0.4	+0.02 −0.03	0.6		0.2		210		0.15
020	5.2		10	−0.12	0.5	+0.02 −0.04	0.75		0.25		325		0.188
027	6.2	+0.12	12.0	−0.20	0.5	+0.02 −0.04	0.85	+0.075 −0.025	0.35		326		0.263
030	6.2		12.5	−0.12	0.7	+0.02 −0.05	1		0.3	+0.075 −0.025	660	+25 −7.5	0.225
150	7.2		14	−0.18	0.8	+0.02 −0.05	1.1	+0.10 −0.05	0.3		797		0.225
040	8.2		16		0.9	+0.03 −0.06	1.25		0.35		1010		0.263
050	8.2	+0.15	20	−0.15	0.6	+0.03 −0.06	1.3	+0.075 −0.025	0.7		412		0.525
060	10.2		20		1.1	+0.03 −0.07	1.55		0.45		1520		0.338
070	10.2		28		1.25	+0.03 −0.07	2.05		0.8	+0.10 −0.05	1850	+15 −7.5	0.6
080	14.2		28	−0.18	1.5	+0.04 −0.08	2.15	+0.10 −0.05	0.65		2840		0.448
160	14.2		28	−0.21	1.0	+0.03 −0.06	1.8		0.8	+0.075 −0.025	1110	+25 −7.5	0.600
090	16.3	+0.18	31.5	−0.18	1.75	+0.04 −0.08	2.45		0.7		3870		0.525
140	16.3		40	−0.25	2.0	+0.04 −0.09	3.1		1.1	+0.10 −0.05	5170		0.825
100	20.4		40	−0.21	2.25	+0.04 −0.10	3.15		0.9	+0.15 −0.05	6500		0.675
170	22.4		45		1.25	+0.03 −0.07	2.85	+0.15 −0.05	1.6		1890		1.200
190	22.4	+0.21	45	−0.25	1.75	+0.04 −0.09	3.05		1.3	+0.10 −0.05	3650	+15 −7.5	0.975
110	25.4		50		3	+0.05 −0.11	4.1		1.1		12000		0.825
120	31		63	−0.21	3.5	+0.05 −0.11	4.9		1.4		15000		1.05
180	36		71		2.0	+0.04 −0.09	4.6	+0.20 −0.05	2.6	+0.15 −0.05	5140		1.950
130	40.5	+0.25	70	−0.30	5.0	+0.05 −0.12	6.2		1.2	+0.10 −0.05	33700		0.900

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AR
Navy - OS

Review activities:
Army - ER
Navy - AS, SH
Air Force - 99
DLA - IS

User activities:
Army - AT, CR, ME
Navy - EC
Air Force - 11

Preparing activity:
Army - AR

Agent:
DLA - IS

(Project 5310-1432)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:57Z
Check the source to verify that this is the current version before use.